

Host Utilities



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**India
SMART UTILITY
Week 2026**
10 – 14 March 2026, New Delhi

Supporting Ministries



सत्यमेव जयते
MINISTRY OF POWER
GOVERNMENT OF INDIA



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CENTRAL ELECTRICITY AUTHORITY

Energy Performance Optimization System (EPOS) *Presented By*

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- **Presentation Time:** The total time for presentation is 12 mins. You are requested to be present on-time as per your session slot.
- **Presentation Format:** All the presenting speakers are required to give a presentation as per the PPT Format. (Maximum 7 Slides). Please note that the additional presentation slides cannot be added in the shared PPT format.

Energy Performance Optimization System (EPOS)

Automating Energy Audit for Distribution Transformers

- ➔ Shift from manual energy audits → real-time intelligent monitoring
- ➔ Meter Data + Analytics + Prescriptive Intelligence
- ➔ 5,000+ Distribution Transformers monitored

Problem in Traditional Energy Audits

➔ Manual data collection

➔ Post-facto loss analysis

➔ No real-time visibility

➔ Delay in fault detection

➔ Time Consuming Process

Why This is Critical

➔ Delayed identification of losses

➔ Regulatory compliance pressures

➔ Higher Operational Cost

➔ Inefficient asset utilization

Why EPOS Matters Today

- Real-time monitoring of **5,259** DT meters
- Integrated DT Health and Energy Accounting Platform
- **Energy Loss Formula:**
$$\text{Loss} = \text{DT Energy} - \Sigma(\text{Consumer Energy})$$
$$\text{Loss(\%)} = \text{Loss Units} / \text{DT Energy}$$
- GIS-based DT-consumer tagging
- Prescriptive Maintenance
- Supports Smart Grid Vision

Strategic Relevance

- ✓ Operational Efficiency
- ✓ Cost Reduction
- ✓ Sustainability
- ✓ Scalable Digital Architecture

1 Data Acquisition

- IoT Smart Meters on DTs
- Voltage, Current, Power Factor

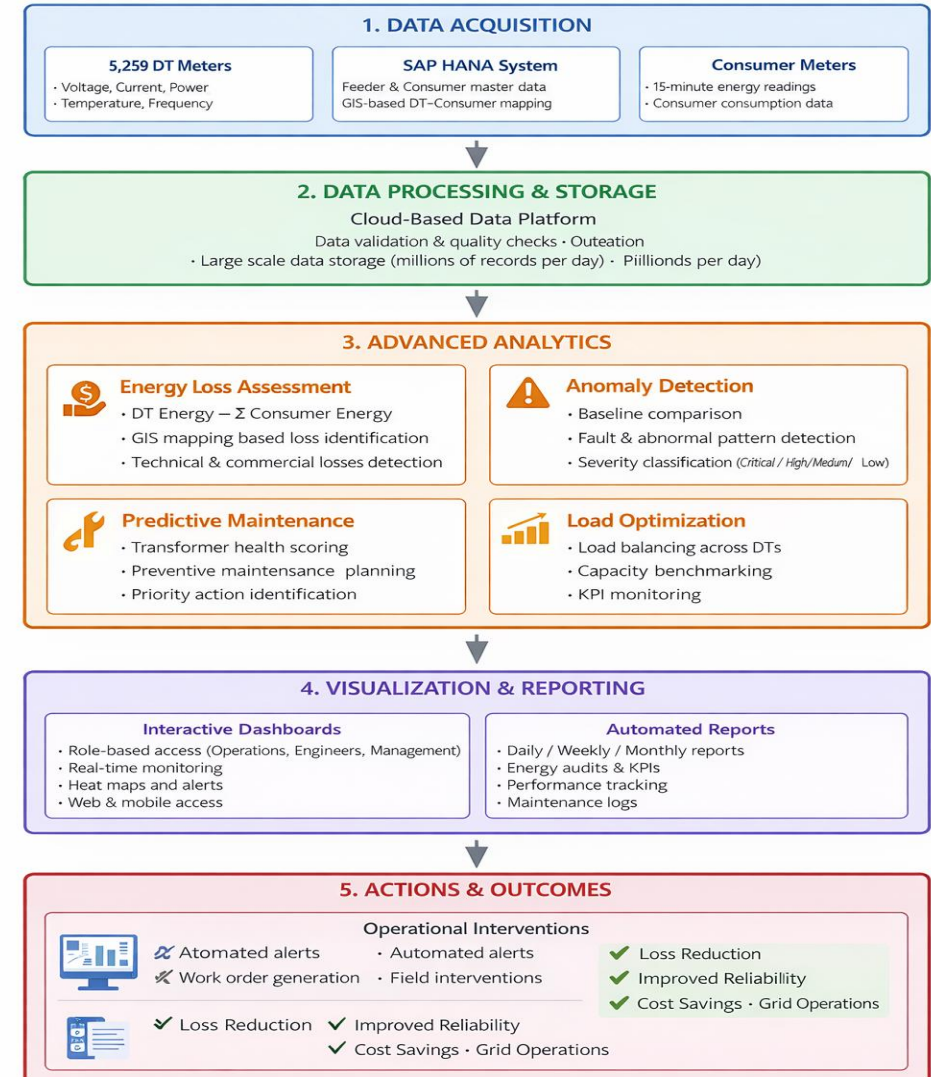
2 Data Pre-processing

- Outlier detection
- Timestamp validation
- 2–3% bad data filtered

3 Analytical Framework

- Load Balancing
- Technical vs Commercial Loss Identification

Smart DT Analytics Framework



Prescriptive Intelligence

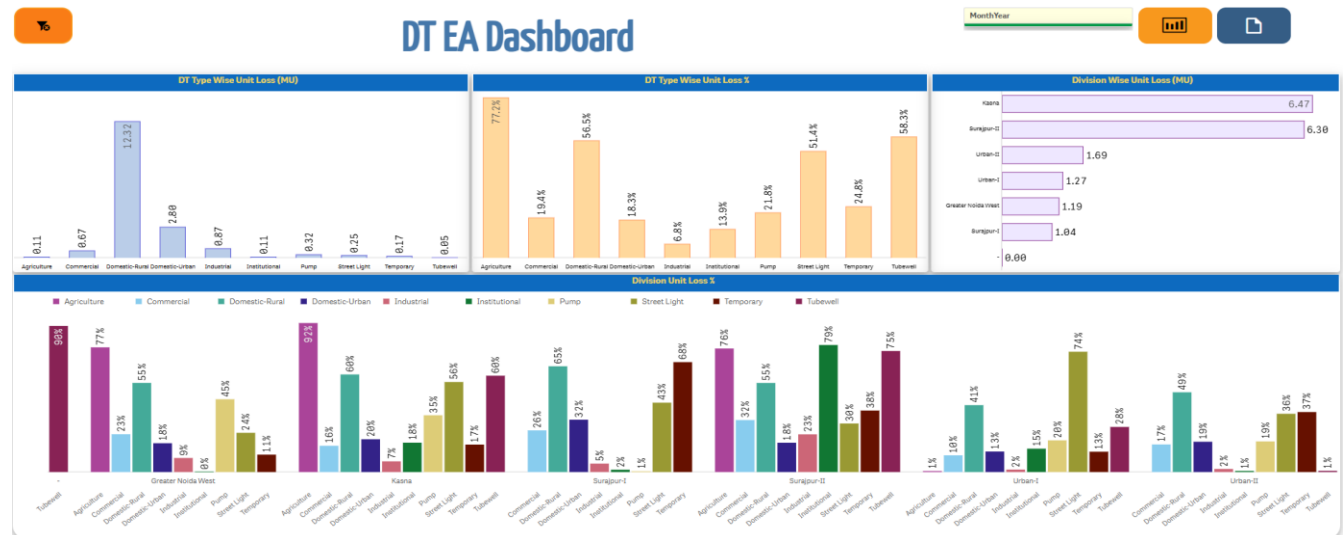
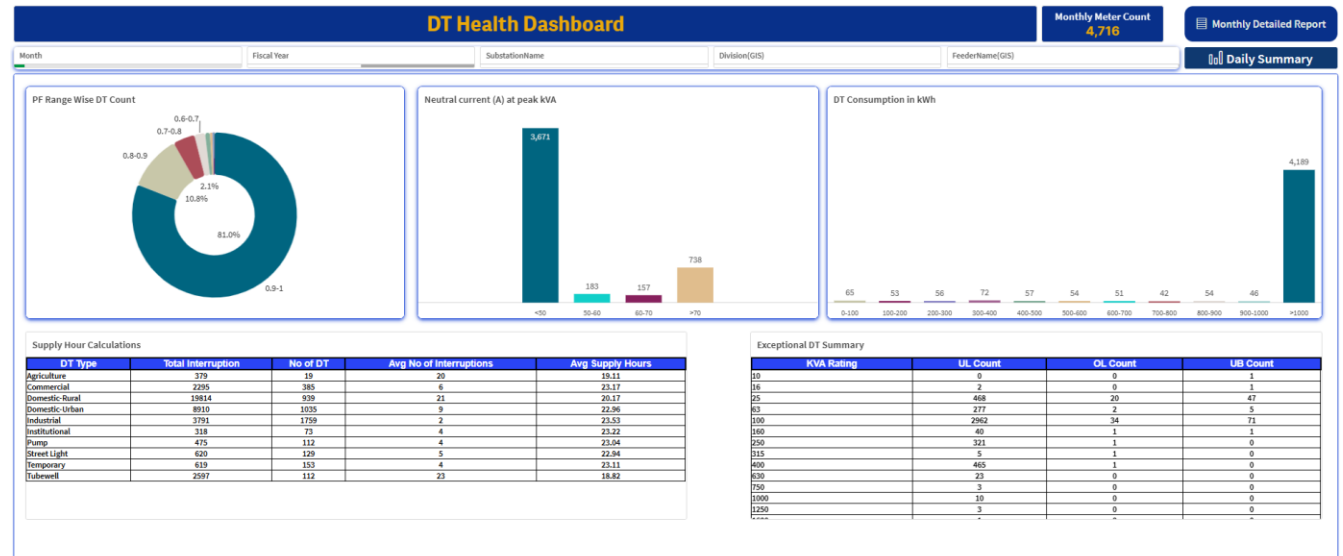
- Pattern recognition from historical failures
- Helps detect potential transformer failures before they occur.

Automated DT Health Indicators

- Power Factor (PF)
- Overloading condition
- Underloading condition
- Load unbalance
- Maximum demand

Visualization Layer

- Role-based dashboards
- Automated daily/monthly reports
- Real-time alerts
- Mobile & web access



Deployment Scale:

- 5,259 Distribution Transformers
- Phased rollout (Pilot → Full Scale)

Outcomes:

- Reduced energy losses
- Improved load optimization
- Faster fault detection
- Reduced downtime
- Better regulatory compliance
- Strong ROI
- Improved Decision and future planning

Sustainability Impact:

- Improves energy efficiency by identifying and reducing DT-level losses
- Supports optimal utilization of existing infrastructure, avoiding unnecessary transformer installations.
- Enables data-driven smart grid operations, supporting long-term sustainable power systems.

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THANK YOU

For discussions/suggestions/queries email: isuw@isuw.in

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Links/References (If any)



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